A PROPER MORPHISM TRIVIAL OFF A FINITE SET OF REGULAR POINTS IS DOMINATED BY A SEQUENCE OF POINT BLOWUPS

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Editorial note. After domination the proof replaces Y by the blowup X'; the source then renames X' to Y for the principalization step.

Allowed standard prerequisites (editorial).

- Dominating modifications by admissible blowups. More on Morphisms of Spaces, lemma dominate-modification-by-blowup (cross-chapter, named). Produces a U-admissible blowup X' -> X dominating Y -> X.
- Principalization of supported ideals. lemma-make-ideal-principal (carded L26829). Turns the defining ideal of the blowup into an invertible one by a point-blowup sequence.
- Universal property of blowing up. Divisors on Spaces, lemma universal-property-blowing-up (cross-chapter, named). Factors the original morphism through the final blowup once the ideal is inverted.

Source reference. Original lemma 0BHG: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 555–555. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 085U: focused-seeds/geometry/S9-spres-dominate-by-blowing-up-in-points/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 087G: focused-seeds/geometry/S9-spres-dominate-by-blowing-up-in-points/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $T \subset |X|$ be a finite set of closed points x such that (1) X is regular at x and (2) the local ring of X at x has dimension 2. Let $f : Y \to X$ be a proper morphism of algebraic spaces which is an isomorphism over $U = X \setminus T$. Then there exists a sequence*

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

where $X_{i+1} \to X_i$ is the blowing up of X_i at a closed point x_i lying above a point of T and a factorization $X_n \to Y \to X$ of the composition.

Proof. By More on Morphisms of Spaces, Lemma 087G there exists a U-admissible blowup $X' \to X$ which dominates $Y \to X$. Hence we may assume there exists an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ such that $\mathcal{O}_X/\mathcal{I}$ is supported on T and such that Y is the blowing up of X in $\mathcal{I}$. By Lemma 0BHG there exists a sequence

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

where $X_{i+1} \to X_i$ is the blowing up of X_i at a closed point x_i lying above a point of T such that $\mathcal{I}\mathcal{O}_{X_n}$ is an invertible ideal sheaf. By the universal property of blowing up (Divisors on Spaces, Lemma 085U) we find the desired factorization. □